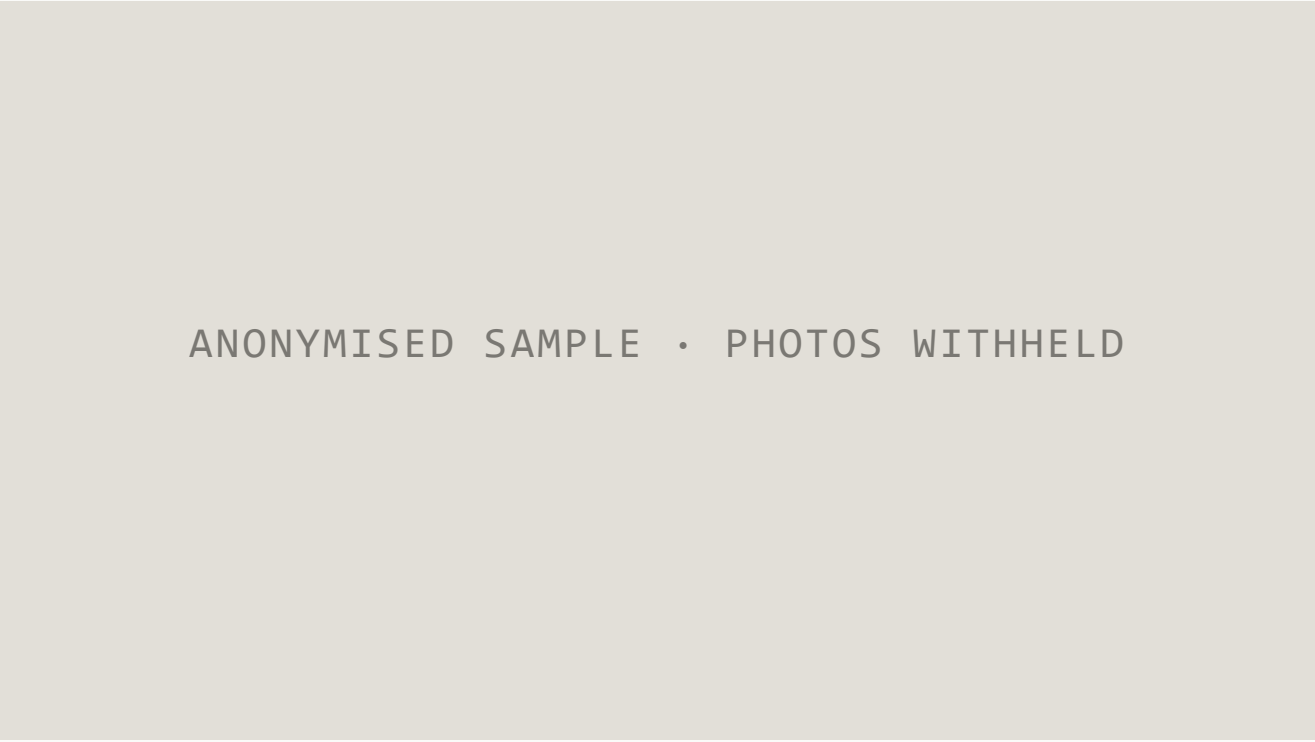
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RETROFIT REPORT · ENERGY, CONDENSATION, VENTILATION AND INSULATION, WITH A PLAN FOR THE WORKS

Anonymised sample report

An assessment of the causes of the high gas consumption, the cold rooms and the recurring condensation and mould, with a costed, staged plan of improvements, from the room-by-room inspection, humidity and moisture readings, extract-fan tests, borescope and drone views and the RdSAP energy assessment made on the day (not lodged as an Energy Performance Certificate, though it can be if you wish), carried out on Thursday 22 October 2026.



PREPARED FOR	VISIT	ISSUED	ASSESSOR
The client	Thursday 22 October 2026, 10.57am–1.00pm	23 October 2026	Dominic Bowkett

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01 At a glance

This report follows my inspection of the property on Thursday 22 October 2026 and responds to your written brief, which asked why the annual gas consumption is so high (sections 07, 10 and 11), whether the house is under-insulated and under-ventilated so that condensation forms (sections 05, 06 and 07), which improvements to prioritise and budget for (sections 12 and 13), and, room by room, why the hall and living room are cold and draughty, why the bathroom and the small rear bedroom keep growing mould, and whether the rear wing is solid brick (section 02). Thank you for the clear brief and the folder of past invoices and guarantees, which settled several questions on the spot.

The house is a detached house of about 1900–1929 with a filled-cavity main building and a two-storey rear wing (the kitchen, bedroom 3, the bathroom and the upstairs hall) built of solid brick about 260 mm thick. The high gas use and the cold, damp rooms have the same few causes: the solid-brick rear wing loses heat at more than twice the rate of the main walls, open chimneys, gappy floorboards and doors let warm air out, the lofts hold only about 100 mm of insulation, and the bathroom fan and the cooker hood do not remove the moisture made in the house. Before any of that is tackled, though, the outside of the house needs attention: missing and slipped roof tiles, the gap in the rear-wing roof, the gutters and the tired paint on the walls and soffits. Repairs come first; improvements follow once the house is weathertight.

Repairs first. Missing and slipped tiles, any leaks into the lofts, the gutters, and the paint and timber on the outside walls, fascias and soffits. Until the house is weathertight, insulation can trap water rather than keep heat in.

Then moisture and draughts. The moisture made in the bathroom and kitchen needs a way out (a working hood, trickle vents, door gaps) before anything is sealed; otherwise draught-proofing and insulation would raise the humidity that is already above the healthy range.

Then insulation, and heating last. The lofts once the roof is sound, then the rear wing and the floor where your builder confirms they can be done safely; the 2021 boiler is sound, and any future heat pump should be sized on the improved house, not on today's.

What I found

01 **The roof and the outside walls need repairs before any improvement work.**

The drone and the ground photographs show slipped and missing tiles at the rear eaves (A28), heavy moss on the north-facing hips of the rear roof (D17–D21), and weathered paint and timber on the render, bargeboards, fascias and soffits (A9, A10). In the rear-wing loft daylight shows where a length of timber appears to be missing (A75; you have seen it too). The roof timbers I could see looked in reasonable condition, but a roofer should check them further, refix the tiles and look for any leaks into the lofts before the lofts are insulated (section 09).

02 **The rear wing is solid brick, as you suspected, and it is the largest single heat loss.**

The kitchen and bedroom 3 walls measure about **260 mm** (tape at the bathroom window reveal, A158), the thickness of a one-brick solid wall, against about 300 mm for the cavity walls of the main building (hall reveal, A179). The energy assessment treats the wing as solid brick as built, for which the method assumes **1.7 W/m²K**, against **0.7 W/m²K** for the filled cavity (RdSAP 10 defaults (BRE, 2026, Tables 6 and 13) [3]). On those assumed figures the wing's 54 m² of wall loses about 92 W/K, more than the 97 m² of main wall (about 68 W/K) and about 35% of the fabric loss of the whole house (section 11, chart C5).

03 **The bathroom and kitchen cannot clear their moisture.**

The bathroom fan extracted **13.1 l/s** on full speed against the **15 l/s** that Approved Document F asks of an intermittent bathroom fan (Testo 417, A172) [1], and there is no trickle vent in the room. The cooker hood gave a reading of about **1.25 l/s** against the 30 l/s required of a hood extracting to outside (A242). A hood is hard to measure consistently (section 06), so treat that figure as a guide, but it is clearly well short. Steam from showers and cooking therefore stays in the house and settles on the coldest surfaces.

04 **The air was above the healthy range in both rooms tested, but the walls themselves were dry.**

I measured **64.9% RH** at 19.0°C in the bathroom (A165) and **64% RH** at 18.8°C in the dining room (A211), against the 40–60% recommended for dwellings [6]. Every surface reading with the Protimeter MMS3 in non-invasive mode fell in the Dry band (highest **162** on the bedroom 3 wall, A144, below the 170 at which the scale turns to At risk [4] [5]). The peeling paint and mould are condensation on cold surfaces, not rising or penetrating damp (section 05).

05 **Open chimneys, gappy floorboards and the doors account for the draughts.**

The assessment counts **four open chimneys**: the living-room fire you use, the open fireplace in the dining room (A205) and the two bedroom flues (A103, A117). The chimneys also ventilate the house, so I would not seal them permanently; your chimney balloons in winter are the right approach, taken out for the rest of the year. The stripped pine floors over the ventilated void show open joints (A178, A213), and the front door and French doors are your own reported draught paths. Fillers between floorboards tend to fall out, so carpet with a thermal underlay in the front living room, and insulation under the floor if the void allows it, are the longer-lasting answers (sections 02 and 07).

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06 **The lofts hold about 100 mm of insulation, much of it compressed under storage boards.**

The main loft is boarded with chipboard over most of its area, with loose mineral wool visible only at the edges (A52, A65, A67); the rear-wing loft has uneven, displaced wool around two redundant water tanks (A71, A77), and the small loft is patchy. The assessment enters 100 mm in both, for which the method assumes 0.4 W/m²K, against 0.16 W/m²K at 270 mm [3] and the 0.16 W/m²K that Approved Document L asks of a renovated roof [2] (section 07).

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07 **The cavity fill is present and dry, but patchy in places.**

The 2005 mineral-fibre fill is documented (CavityTech invoice and the CIGA 25-year guarantee to August 2030, A243–A245). Through the borescope holes in each elevation I saw fibre filling most of the cavity, with some bare brick faces, mortar snots and small voids (B6, B28), and no wetness. The fill is doing its job; it does not need extraction. Ask CIGA to record the guarantee now, because you believe the installer was dissolved in September 2024 (section 07).

---

08 **The suspended timber floor is uninsulated over a ventilated void.**

Under the lifted board in the dining room the joists, pipes and void are visible with no insulation (A213, A214). The method assumes a floor U-value of 0.83 W/m²K for it, against 0.25 W/m²K for a renovated floor [2]. The void needs its airflow, so the air bricks must never be blocked; whether the void is deep enough to insulate from below is for an insulation contractor to confirm (sections 02 and 07).

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09 **The boiler is new enough and well documented; the controls are good but three radiators lack valves.**

A Worcester Greenstar 32CDi Compact ErP condensing combi (GC 47-406-79, A239) fitted in 2021, serviced annually and guaranteed to 2031 (your brief). Hive provides the programmer and room thermostat (A231), and TRVs are on five of the nine radiators; the living room, dining room and kitchen have none (section 10).

The five things I would do first

#	ACTION	INDICATIVE COST
1	**Have a roofer refix the slipped and missing tiles, check the lofts for leaks and check the roof timbers**, including the missing length in the rear-wing loft, and clear the gutters – Stage 0, repairs (section 13).	Obtain a quote; gutters £150–£230 [24]
2	**Repair and repaint the outside walls, fascias, soffits and bargeboards** with a breathable masonry paint in the spring redecoration you already plan – a decorator, Stage 0.	Obtain a quote
3	**Replace the cooker hood with one ducted to outside** at 30 l/s or more, and use it on boost whenever the hob is on – an electrician, Stage 1.	£200–£400 plus the hood [19]
4	**Fit trickle vents to every window that has none** (all except bedroom 3's) and trim the three doors that are short of a 10 mm gap – a window installer and a carpenter, Stage 1.	£260–£520 [20]
5	**Top up the lofts to 270 mm** on raised storage boards, with the hatches insulated and draught-stripped, once the roofer has confirmed the roof is sound – an insulation installer, Stage 2.	£950–£2,350 [12]

Costs are indicative 2026 ranges from the sources listed in section 17, regional where a figure for the area was published and national otherwise, scaled to this house for budgeting. They are a guide only: actual prices vary with the contractor, access, what is found once work starts, and changes in prices. Obtain three written quotes before committing.

02 Your questions, answered

“Why is our annual gas consumption so high?”

Because a good share of the house is poorly insulated and leaky, and the heating now runs for longer hours than it used to. Your bills show **16,961 kWh** of gas from 1 July 2025 to 20 June 2026, about 1.8 times Ofgem's typical medium value of 9,500 kWh (16,961 ÷ 9,500 = 1.79) [7]. On the method's assumed U-values the solid-brick rear wing alone is about 35% of the fabric heat loss (finding 02); the open chimneys and the gappy floors add air leakage the assessment can count but not measure; the lofts are at about 100 mm; and, since the damp specialist's visit, you have kept the heating on more constantly at about 18–19°C, which is sensible for the mould but adds hours of running. The gas hob, used until May 2026, accounts for a small part of the total. **What to do:** the repairs first (Stage 0), then the moisture, draught and loft measures (Stages 1 and 2), and the rear-wing and floor insulation once your builder has advised on them (Stage 3, section 13).

“Was the kitchen extension built with cavity walls, or is the small rear bedroom solid brick only?”

Solid brick. The kitchen and bedroom 3 walls measure about 260 mm, which is one brick plus plaster, and the energy assessment records the whole rear wing as solid brick on both floors, as your floor-plan notes did. The main building is a 300 mm cavity wall, filled in 2005. The age of the rear wing is not known, so the assessment gives it the same age as the main house (1900–1929). **What to do:** consider insulating the wing's walls, inside or outside, with your builder's advice (section 07, Stage 3).

“Would insulated lining paper or insulated plasterboard on the small rear bedroom's outside walls help, as the previous surveyor suggested?”

It would help, but it is not a simple job, and it is something to think about rather than rush. The previous advice and mine agree on the cause: the walls are cold, so moisture condenses on them. A thin insulated lining paper raises the surface temperature slightly; an insulated plasterboard of about 50 mm brings the wall close to the 0.30 W/m²K that Approved Document L asks of an insulated wall [2]. But internal insulation on a solid wall carries real risks: the brick behind it becomes colder and wetter, any gap in the vapour control layer lets moist room air condense behind the board, and the junctions with the floor, ceiling and window reveals stay cold [27]. Those risks are highest in rooms that make a lot of moisture, such as a bathroom or kitchen. Bedroom 3 is also small (4.67 m²), so the lost thickness matters. The alternative is to insulate the wing from outside (section 07). Which is more suitable can only really be settled with a builder who has looked at the walls, and either way the ventilation has to be put right first. **What to do:** discuss the options in section 07 with your builder before the bathroom refurbishment is priced (Stage 3).

“Are the gaps between the floorboards essential for ventilation, and would we be sealing up too much if we filled them or insulated underneath?”

No, the gaps are not needed; the air bricks are. The void under the floor is ventilated through the air bricks in the outside walls (A4, A38), and it is that airflow that keeps the joists dry. The gaps between the boards only let cold void air into the rooms, so closing them does not reduce the ventilation the floor needs, as long as every air brick stays clear. In practice, though, sealing the gaps is difficult: rubber strips and fillers tend to work loose and fall out as the boards move, so treat it as a temporary measure. Longer-lasting options are carpet with a thermal underlay, which would suit the front living room, and insulation between the joists if the void allows it. The void is a small space and I could not tell whether it is deep enough to insulate from below; an insulation contractor can confirm that. **What to do:** keep the air bricks clear, consider carpet and thermal underlay in the living room, and ask an insulation contractor about insulating the floor (section 07, Stage 3).

“Is there mould behind the bathroom tiles?”

Probably not on any scale. The pinless meter read **142** on the tiles by the bath and **150** on the wall beside the window (A166, A168), both in the Dry band [5]. The yellow and black mould is on the surface of the grout, the shower curtain and the ceiling paint, where steam condenses. The tiles will come off in the refurbishment, so look then; anything more than surface staining on the plaster should be cut out. **What to do:** a trickle vent now (Stage 1), a better fan in the refurbishment (section 06, Stage 3), and a mould wash meanwhile.

“Why is the hall still cold and draughty after the new front door?”

Because the door was one of several paths. The hall floor is bare boards over the ventilated void (A178), the living-room flue draws air through the house whenever it is open, and your paper test showed the front door closing loosely on its seal. The hall faces north-east. **What to do:** adjust the front-door keeps and seal (the installer's guarantee runs to 2031), use your chimney balloon in the living-room flue in winter when the fire is not in use, and consider a runner or rug on the hall boards (Stage 1).

“Why is there a window in the former tank cupboard?”

The former tank cupboard is the closet downstairs in the main part of the house, beside the dining room. Its small “window” does not open to the outside: it is a glazed panel looking into a void packed with loose insulation fibre, which is the fluff you can see behind the glass (A209, A210). It was most likely left from when the tank stood there, perhaps as an access or inspection opening that was later glazed over. It does no harm and needs no work; leave the glass in place so the fibre stays where it is.

“The loft insulation looks old and amateur, things stored there smell musty, and a roof timber over the rear bedroom seems to be missing. What should we do?”

Have the roof checked and repaired first, then top up the insulation and raise the storage boards. The insulation is old mineral wool, compressed under the chipboard (A52) and displaced in the rear-wing loft (A71, A77). The timbers I could see looked in reasonable condition, but a roofer should check them further, including the missing length in the rear-wing loft (A75), and look for any leaks while they refix the slipped and missing tiles. The musty smell is consistent with a cold, slightly damp roof space: there is no roofing felt under the tiles, only the original boarding (A57, A68), so moist air rising through the hatches condenses on the cold timber. **What to do:** the roofer first (section 09, Stage 0), then the loft top-up with hatch seals (section 07, Stage 2). Store soft items in sealed boxes in the meantime.

“Which improvements should we prioritise, and how should we budget for them?”

Repairs first, improvements after. Start with the defects: the roof tiles, any leaks into the lofts, the gutters and the outside paintwork (Stage 0). Then the moisture and draught measures, which cost about £880–£1,680 together (Stage 1), then the lofts (Stage 2), and the rear wing, bathroom and floor over the next three years, timed to your bathroom refurbishment and decided with your builder (Stage 3). The order and costs are set out in section 13. **What to do:** book the roofer and the decorator (Stage 0).

“Are the cracks in the living-room ceiling and walls a concern?”

I do not see them as a major concern. Hairline cracks like those in the living room, bedroom 1 (A99), the kitchen ceiling (A240) and the bathroom ceiling (A159) are typical of old plaster moving with the seasons. They should still be managed and monitored: note their length and width, fill and redecorate, and watch whether they reopen. If any crack widens, keeps reopening after filling, or makes a door stick, ask a structural engineer (section 16); this report is not a structural assessment.

Decisions you raised on the day

The bathroom refurbishment, early 2027

You suggested: moving the extract fan to directly above the shower; your builder visited the day before mine.

My advice: yes, put the fan high on the wall or in the ceiling close to the shower and as far from the door as possible, ducted to outside, running at 8 l/s continuously with a 15 l/s boost or more (Approved Document F, Tables 1.1 and 1.2 [1]); fit a trickle vent of 4,000 mm² and undercut the door by 10 mm. Whether to insulate the two outside walls at the same time is a question for your builder: internal insulation in a wet room carries the risks set out in section 07, and external insulation of the wing is the alternative.

What it affects: the bathroom mould, bedroom 3's damp, the rear-wing heat loss.

Decision needed by: before the builder prices the work (Stage 3).

Painting the outside next spring

You suggested: external redecoration in spring 2027 (the decorator's quote is still to come).

My advice: treat this as a priority repair, not decoration: repair the render cracks (A9), repaint the walls with a breathable masonry paint, and repair and paint the fascias, soffits and bargeboards, replacing any soft timber. If a scaffold goes up, use it to refix the slipped tiles, clear the gutters and treat the moss.

What it affects: keeping the walls and roof dry before any insulation work.

Decision needed by: before the decorator is booked (Stage 0).

Bedroom 3's trickle vent

You suggested: your daughter may close it when she is home because the room feels cold.

My advice: keep it open; the room is cold because of its walls, not the vent, and closing it raises the condensation on those walls.

What it affects: the mould in bedroom 3.

Decision needed by: now.

03 The house and how you use it

Type and age	Detached two-storey house with a two-storey rear wing, age band 1900–1929 for both parts in the energy assessment (the rear wing's age is not known, so it is given the same age as the main house); No 44 next door carries a 1910 plaque (pre-visit desk research). Past works from your brief and the papers photographed: injected damp-proof course and timber treatment 1987 (guarantee expired); possible wet-rot treatment to the living room 1995; cavity fill 2005 (A243–A245); rear guttering and a downpipe replaced 2016 (A246); bathroom fan 2016; bedroom 3 window 2016 (A247); dining-room and under-stairs damp-proof course 2018; roof tiles, stack flashing and chimney caps 2015/2018; front door and combi boiler 2021 (guaranteed to 2031); kitchen and French-door glass 2024 (to 2029).
Floor area	85.6 m² internal: main building 33.19 m² on each floor, rear wing 9.60 m² on each floor; room heights 2.41–2.45 m (measured on the day; the same figures as the assessment entries).

Walls	Main building: cavity wall about 300 mm overall, brick ground floor and roughcast render above, filled with mineral fibre in 2005, CIGA guarantee to August 2030 (A179, A245, borescope B21–B31). Rear wing: one-brick solid wall about 260 mm, brick below and render above, uninsulated (A158).
Floors	Main building: suspended timber, stripped pine boards over a ventilated void, uninsulated (A213, A214); air bricks in the side and front walls (A4, A38). Rear wing: solid, laminate finish in the kitchen; construction not seen (floor-plan notes).
Roof and loft	Pitched, plain clay tiles on timber boarding with no felt (A57, D13). Three loft spaces: main (hatch in bedroom 1, largely boarded for storage), rear wing (hatch in bedroom 3, two redundant tanks) and a small loft (hatch only). About 100 mm of mineral wool at the joists in each (entered in the assessment).
Windows and doors	Nine white uPVC double-glazed units, most from before your purchase in 2001, entered as date unknown with a 16 mm or larger gap; bedroom 3 replaced in 2016 with 28 mm soft-coat low-E units and a trickle vent (A247). Only bedroom 3's window has a trickle vent. Front door 2021; uPVC back door and French doors. No FENSA certificate was found for this address in the pre-visit search (21 October 2026); the 2016 window may have been certified through another scheme.
Heating and hot water	Worcester Greenstar 32CDi Compact ErP condensing combi, GC 47-406-79, 89.2% efficient (product database entry used in the assessment), in a kitchen wall cupboard (A236, A239); radiators; Hive receiver (A231) and Hive thermostat; TRVs on bedrooms 1–3, the landing and the hall (5 of 9 radiators), none in the living room, dining room or kitchen (your list). Hot water from the combi. Open fire in the living room in use (A191); an open fireplace in the dining room, rarely used.
Ventilation	Natural, with an intermittent bathroom fan (2016, A162) and a chimney cooker hood (A225); four open flues; a trickle vent in bedroom 3 only (section 06).
Meters, tariff and bills	Mechanical gas meter, **45009** cubic feet on the day (A217); Aclara SGM1312 smart electricity meter, **42523 kWh** import on the day (A218). Octopus, on the price-capped tariff since a fixed rate ended in August 2026. Gas 16,961 kWh (1 July 2025 – 20 June 2026), about £1,405 a year; electricity 2,605 kWh (1 October 2025 – 30 September 2026), about £1,090 a year (your brief). Hob and oven all-electric since May 2026.

Setting and orientation	Suburban street; the front faces about north-east onto the road and the garden south-west (pre-visit desk research); side passages on both sides (A11, A39); a garden room at the rear (A22). The main roof's ridge runs front to back, so its slopes face south-east and north-west.
Designations and planning	No conservation area, listing, Article 4 direction or tree order at the house; the nearest conservation area (a nearby main road) is about 130 m away. No planning applications for the house or its neighbour (pre-visit desk search of the planning register, 21 October 2026); none is likely to be needed for the works in this plan except, possibly, external insulation of the rear wing, which changes its appearance; I cannot confirm that position, so check it with the local council planning team or the Planning Portal before committing [28].

PHOTO WITHHELD FOR PRIVACY

Front elevation, north-east. Brick ground floor, roughcast first floor and mock-Tudor gable; the main building is the filled-cavity part.
A2 · 12:32

PHOTO WITHHELD FOR PRIVACY

Rear, south-west. The two-storey rear wing (right of centre) is solid brick; the French doors are in the main building.
A26 · 12:39



Living room. Open fire in use and stripped boards with open joints; no TRV and no trickle vent.
A181 · 12:03



Dining room. Open fireplace, radiator without a TRV, French doors to the garden.
A196 · 12:07



Kitchen (rear wing). Solid-brick walls; the windows steam up when cooking (your brief).
A215 · 12:05



Bedroom 1 (front). Main loft hatch above; the wardrobe on the outside wall smells musty; no trickle vent.
A94 · 12:01



Bedroom 2 (middle). No trickle vent; condensation on the lower panes in winter (your brief).
A109 · 11:59



Bedroom 3 (rear wing). Three outside walls of solid brick; the 2016 window has a trickle vent.
A135 · 11:56

How you use the house

Four adults make up the household: you, your husband and your twin son and daughter, aged 20, who are at university and home in the holidays. For much of the year the house is occupied by two people, which is the count I recorded on site, so the moisture load and the heating pattern are those of two people, rising to four in the holidays when bedrooms 1 and 3 are in use. The energy assessment assumes a standard occupancy for a house of this size regardless. The heating runs on a Hive schedule at about 18–19°C, more constantly since the damp specialist's advice a couple of years ago, which you found helped. Showers are taken with the fan on and the window open until the steam clears; laundry is sometimes dried indoors and otherwise in a tumble dryer in the garden room; the living-room fire is lit occasionally, the dining-room one rarely, the bedroom flues never. You bought the house in 2001 and expect to stay, so the plan looks ten years ahead; your own plans and the triggers they set are in the table in section 13.

Evidence gathered on the day. Time on site about 10.57am to 1.00pm, starting with our conversation and followed by the inspection; 249 room and external photographs; 2 relative-humidity and temperature points (Protimeter MMS3 hygrometer); 8 moisture-meter points (Protimeter MMS3, non-invasive mode); 2 extract tests (Testo 417 anemometer); door undercuts measured with a rule; 31 borescope frames through drilled mortar joints in two elevations, made good afterwards; 25 drone frames of the roofs and elevations; documents reviewed: your two-page brief, the cavity-fill invoice and CIGA guarantee, the 2016 guttering and window invoices, the boiler service history; the energy assessment made on the day. Weather: rain in the early morning cleared before the visit; dry during the inspection, about 12–13°C outside with 66–75% relative humidity and a light breeze, after an overnight low of about 7°C (modelled local data) [26].

04 Floor plan and where the problems are

Internal dimensions in metres, to the inside face of the external walls; h is the floor-to-ceiling height. Red lines are the external walls (the heat-loss perimeter); grey dashed lines divide the main building from the rear wing; thin lines are room runs and internal walls; blue bars are windows (W numbers, including the French doors) and orange bars the solid external doors; red bars are radiators, labelled TRV or No TRV; HS marks the boiler, T the thermostat and P the programmer; EM and GAS the meters; dashed brown boxes the loft hatches; the yellow-shaded part marked Ext 1 is the solid-brick rear wing; north as arrowed. Extract fans are not drawn (see section 06). The numbered circles mark the findings listed below.

GROUND FLOOR



FIRST FLOOR



1. **Kitchen (rear wing).** Cooker hood about 1.25 l/s against 30 l/s; no trickle vent; solid-brick walls about 260 mm.
2. **Dining room.** 64% RH; peeling paint by the radiator, 152, Dry; uninsulated floor void; open fireplace; no TRV; no trickle vent in the French doors.
3. **Hall.** Cold and draughty: front door seal, bare boards over the void, stair to the landing.
4. **Living room.** Open fire in use; floorboard gaps; no TRV; no trickle vent.
5. **Bedroom 3 (rear wing).** Solid walls; peeling paint by the radiator; wall 162, Dry; loft hatch to the rear-wing loft.
6. **Bathroom.** Fan 13.1 l/s against 15 l/s; 64.9% RH; mould on grout and ceiling; no trickle vent.
7. **Bedroom 2.** Open fireplace; no trickle vent; door undercut below 10 mm; cupboard window.
8. **Bedroom 1.** Undercut 7 mm; no trickle vent; fireplace; musty wardrobe on the outside wall; main loft hatch.

05 Moisture, condensation and mould

BATHROOM HUMIDITY

64.9%

at 19.0°C, 12.18 (A165)

DINING-ROOM HUMIDITY

64.0%

at 18.8°C, 12.28 (A211)

HEALTHY RANGE

40–60%

relative humidity [6]

CONDENSATION POINT

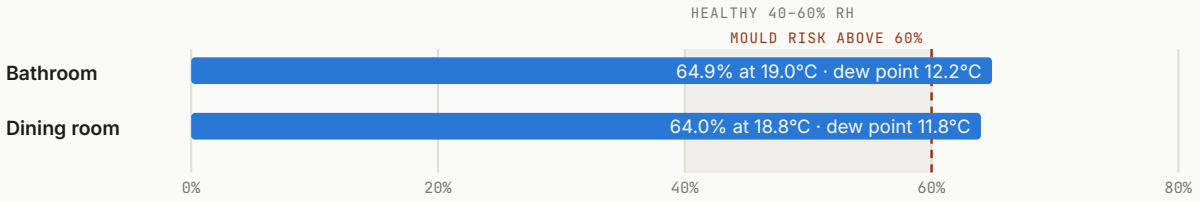
~12°C

surfaces colder than this get wet (bathroom reading)

Humidity

I measured the air with the Protimeter MMS3 hygrometer in the two rooms where you see the most condensation and peeling paint. The bathroom read **64.9%** at 19.0°C at 12.18, just before the fan test there; the dining room read **64%** at 18.8°C at 12.28. Both are above the 40–60% range recommended for dwellings in the building-services guidance (CIBSE TM40) [6]. The heating had been running on its usual schedule and the day was dry enough for the drone to fly; both tend to lower the readings, so on a wet winter evening with the windows shut and laundry drying the figures will be higher. Outside, the morning's rain had cleared and the air was at about 66–75% relative humidity and 12–13°C during the visit [26].

The figure that matters for mould is the dew point: at 64.9% and 19.0°C the bathroom air reaches its dew point at about **12.2°C**, so any surface colder than that, such as the window glass, the cold corner of the solid outside wall or the ceiling under a thin layer of loft insulation, will become wet. In the dining room the dew point is about 11.8°C. On a 5°C night the inner face of an uninsulated solid-brick wall can fall close to these temperatures, which is why bedroom 3 and the bathroom, both in the solid rear wing, have the worst of the mould.



Relative humidity by room, with the dew point of each reading. Protimeter hygrometer readings on Thursday 22 October 2026 (A165, A211); the shaded band is the 40–60% recommended for dwellings in CIBSE TM40:2020, as quoted by [6]. Two readings only, taken around midday; they show the level on the day, not the winter peak.

Readings

LOCATION	READING	INSTRUMENT	WHAT IT MEANS	RESULT
Bedroom 3, wall beside the window	**162**	Protimeter, non-invasive (A144)	Highest reading of the day; dry, near the top of the band, consistent with a cold wall that dries between events.	DRY
Bedroom 3, wall near the window	**153**	Protimeter, non-invasive (A147)	Dry.	DRY
Bedroom 3, flaking plaster above the radiator valve	**152**	Protimeter, non-invasive (A148)	Dry where the paint is failing: the damage is from past condensation, not water in the wall.	DRY

Bathroom, tiles by the bath	**142**	Protimeter, non-invasive (A166)	Dry behind the tiles at the point tested.	DRY
Bathroom, window sill	**86**	Protimeter, non-invasive (A167)	Dry.	DRY
Bathroom, wall beside the window	**150**	Protimeter, non-invasive (A168)	Dry.	DRY
Bathroom, ceiling by the peeling paint	**129**	Protimeter, non-invasive (A170)	Dry; the flaking is from repeated condensation.	DRY
Dining room, wall by the radiator	**152**	Protimeter, non-invasive (A212)	Dry where the paint peels; the 2018 damp-proof course appears to be holding at this point.	DRY
Bathroom air	**64.9% RH**, 19.0°C	Protimeter hygrometer (A165)	Above the healthy range; dew point about 12.2°C.	BORDERLINE
Dining-room air	**64% RH**, 18.8°C	Protimeter hygrometer (A211)	Above the healthy range; dew point about 11.8°C.	BORDERLINE

Protimeter MMS3 in non-invasive (search) mode on its 60–999 relative scale [4]: readings are comparative, not a moisture content. DRY 60–169; BORDERLINE 170–199 (Protimeter's "At risk"); WET 200–999, as Protimeter publishes for this scale [5]. For air, Borderline means above the 60% upper limit of the healthy range [6].



Bedroom 3, beside the radiator: vertical strips of failed paint at the corner of two solid outside walls, the coldest point of the room.
A139 · 11:57



Bedroom 3, ceiling corner: stained, flaking paint where the cold wall meets the ceiling under the thin loft insulation.
A138 · 11:57



Bedroom 3, flaking plaster above the valve: 152 on the meter, Dry.
A148 · 12:19



Bathroom ceiling: mould specks around the downlight and pull cord, where steam rises and meets the coldest surface.
A160 · 11:58



Bathroom, above the window: cracked and peeling plaster on the solid outside wall.
A161 · 11:58



Bathroom window: condensation on the glass at midday, with the window open earlier.
A156 · 11:58



Bathroom air: 64.9% RH at 19.0°C.
A165 · 12:18



Dining-room air: 64% RH at 18.8°C.
A211 · 12:28



Dining room, wall by the radiator: peeling paint, 152 on the meter, Dry.
A212 · 12:28

Why it is happening

- **Too much moisture stays in the house.** Four people in the holidays, showers, cooking and indoor drying produce water vapour every day; the bathroom fan removes it at 13.1 l/s rather than 15 l/s and the hood hardly at all (section 06).
- **The rear wing has the coldest walls.** One brick thick and uninsulated, bedroom 3's walls and the bathroom's run several degrees colder than the main building's filled cavity walls, so they reach the dew point first.
- **Most rooms have no fresh-air path.** Only bedroom 3's window has a trickle vent, and several doors are undercut less than 10 mm, so moist air cannot move to the extract points.
- **Storage limits air movement.** Full cupboards and wardrobes on outside walls (bedroom 1, the under-stairs cupboard, the kitchen base units) stay cold and still, which is why clothes and jackets there smell musty or grow white mould.

The meter readings rule out the alternatives: no surface tested read above the Dry band, the walls are not wet behind the paint, and at the points tested the 1987 and 2018 damp-proof courses are holding.

What to do now

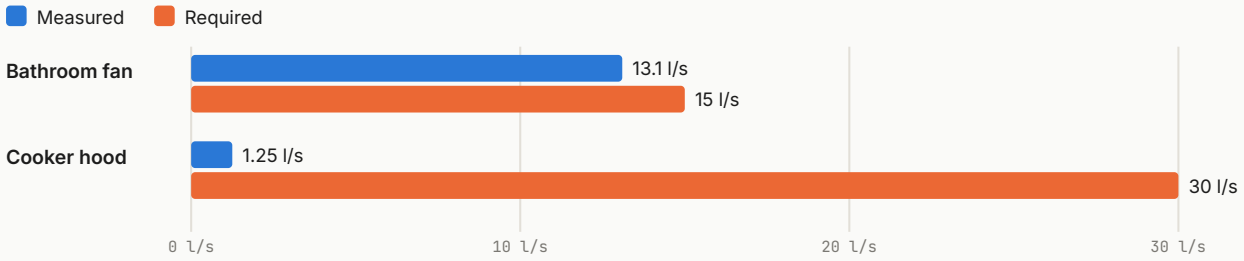
- Run the bathroom fan for 20 minutes after every shower with the door shut and the window closed, so the fan draws air from the house rather than from the window; wipe the tiles and glass down.
- Use the hood on its highest setting with lids on pans; open the kitchen window a little while cooking until the new hood is in.
- Keep the trickle vent in bedroom 3 open, and the new ones once they are fitted.
- Dry laundry outside or in the tumble dryer whenever you can; if indoors, in the bathroom with the fan running and the door shut.
- Leave a small gap between furniture and the outside walls, and do not pack the wardrobes on outside walls tight.
- Treat existing mould with a fungicidal wash and repaint with a mould-resistant emulsion only after the ventilation is improved.
- Buy a digital hygrometer for the landing and bedroom 3, and aim to keep below 60% (section 13).

06 Ventilation

ROOM	BACKGROUND VENTILATION (PASS / FAIL)	EXTRACT (MEASURED / REQUIRED)	DOOR UNDERCUT (PASS / FAIL)	RESULT
Kitchen	None – **Fail**	Hood about **1.25 l/s** / 30 l/s (A242; indicative, see below)	**Pass**	FAIL
Bathroom	None – **Fail**; the window is opened during showers	Fan **13.1 l/s** / 15 l/s (A172)	**Fail** – insufficient	FAIL
Living room	None – **Fail**	Not required	**Pass**	FAIL
Dining room	None in the French doors – **Fail**; open fireplace	Not required	**Pass**	PLAN
Hall	None (circulation space)	Not required	–	PASS
Bedroom 1	None – **Fail**	Not required	**Fail** – **7 mm** against 10 mm (measured on the day, A105)	FAIL

Bedroom 2	None in the window or the cupboard window – **Fail**	Not required	**Fail** – below 10 mm	FAIL
Bedroom 3	One trickle vent, 150 mm long – **Pass**	Not required	**Pass** (A142)	PASS
Landing	None (circulation space)	Not required	–	PASS

Method: extract rates measured with a Testo 417 vane anemometer at the fan and hood grilles, on full speed. A cooker hood is difficult to measure consistently: its intake is wide and covered by grease filters, the air moves unevenly and turbulently across it, and the reading changes with where the anemometer is held, so the hood figure is a guide to its order of magnitude rather than a precise rate. Approved Document F Volume 1 (2021) [1]: Table 1.1 asks 15 l/s of an intermittent bathroom fan, 30 l/s of a kitchen cooker hood extracting to outside, or 60 l/s of a kitchen fan elsewhere; Table 1.2 asks 8 l/s of a continuous bathroom fan and 13 l/s high rate in a kitchen; Table 1.7 asks background ventilators of 8,000 mm² equivalent area in each habitable room and kitchen of a house of more than one storey, and 4,000 mm² in a bathroom; paragraph 1.25 asks the equivalent of a 10 mm undercut on a 760 mm door (7,600 mm²), so air can reach the extract fans with the doors shut. For a three-bedroom house, Table 1.3 sets a whole-house rate of 31 l/s. For an existing home the works must leave the ventilation no worse than before (paragraph 3.15); the minimums are the standard a new window or fan should meet. PASS meets the rate; FAIL falls short of it; PLAN falls short but is dealt with at a later stage.



Extract measured against the rate required. Testo 417 readings on Thursday 22 October 2026 (bathroom A172, hood A242); required rates from Approved Document F Volume 1, Table 1.1 [1], taking the hood as extracting to outside (if it recirculates, a 60 l/s kitchen fan is needed instead). The bathroom fan reaches 87% of its rate (13.11 ÷ 15); the hood reading is indicative only, but far short of 30 l/s.



Door undercut measured against the 10 mm guidance. Bedroom 1 measured with a rule on the day (A105); the 10 mm line is Approved Document F paragraph 1.25 [1]. A 7 mm gap gives 5,320 mm² on a 760 mm door, 70% of the 7,600 mm² intended. Bedroom 2 and the bathroom were also short of 10 mm on the day (Fail in the table); only bedroom 1 was measured.



Bathroom fan: Airflow wall fan fitted in 2016, on the light switch with a 20-minute overrun.
A162 · 11:58



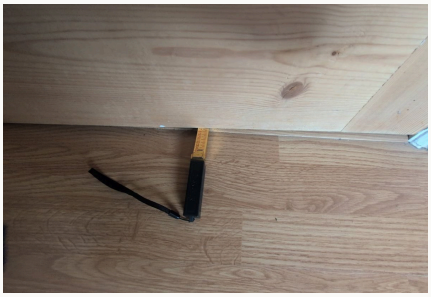
Bathroom fan test: 2.05 m/s, 13.11 l/s on full speed, against 15 l/s; a first reading a minute earlier gave 0.51 m/s, 2.83 l/s (A171) before the meter settled on the grille.
A172 · 12:20



Cooker hood: stainless chimney hood over the range, old and greasy.
A225 · 12:06



Hood test: about 1.25 l/s at the grille, against 30 l/s; an indicative reading.
A242 · 12:13



Bedroom 1 door: the rule in the gap; 7 mm against 10 mm.
A105 · 12:01



Bedroom 3 door: a sufficient undercut over the new floor.
A142 · 11:57

The options

SYSTEM	HOW IT WORKS	FIT FOR THIS HOUSE
Intermittent fans	Run when the light or a humidistat switches them on; draw replacement air through trickle vents and undercuts.	What you have; adequate only if each fan meets 15 l/s (bathroom) or 30 l/s (hood) and the room has a vent.
dMEV (decentralised mechanical extract ventilation)	A quiet fan in each wet room that runs continuously at a low rate and boosts when humidity rises.	**Recommended** for the bathroom in the refurbishment: it clears the moisture between showers as well as during them, and suits a small room with a cold wall.
PIV (positive input ventilation)	A loft unit that blows filtered air into the landing, pushing moist air out through leaks.	Not recommended: it pushes moist air into the cold rear-wing walls and through the unfelted roof, and adds heat load.

Central MEV or MVHR	Ducted whole-house extract, with heat recovery in MVHR.	Not proportionate: the house is not airtight enough for MVHR to pay back, and the ducting would cross the boarded lofts.

What I recommend

- **Cooker hood (Stage 1).** A chimney hood at least the width of the range, ducted to outside through the rear-wing wall in 125 mm or 150 mm rigid duct with a back-draught shutter, rated at 30 l/s or more at the setting you use, 650–750 mm above the hob unless the maker says otherwise (Approved Document F paragraph 1.21) [1]. £200–£400 plus the hood [19].
- **Bathroom fan (Stage 3, in the refurbishment).** A dMEV unit high on the wall or in the ceiling near the shower, 8 l/s continuous with a humidity-controlled boost of 15 l/s or more, rigid duct to an external wall grille, on an isolator outside the room, the electrical work notified under Part P of the Building Regulations; commissioned with a flow test. £250–£450 [18]. Until then, the existing fan with the door closed.
- **Trickle vents (Stage 1).** In every window that has none – the living room, bedroom 1, bedroom 2, the kitchen and the bathroom – marked with their equivalent area: at least 8,000 mm² in each habitable room and 4,000 mm² in the bathroom, split across the frames where one vent will not reach it. £200–£400 [20].
- **Door undercuts (Stage 1).** Trim bedroom 1, bedroom 2 and the bathroom doors to 10 mm above the floor finish; a carpenter can do all three in a visit. I estimate £20–£40 a door, £60–£120 for the three. The French doors in the dining room can take a vent in the frame, or one when they are next replaced.

07 Insulation

The lofts

Three roof spaces sit over the bedrooms. The main loft, through the hatch in bedroom 1, is boarded with chipboard over most of its area for storage; loose mineral wool shows only at the board edges and the eaves, where the tape put it at about 100 mm, compressed in places (A52, A65, A67). The rear-wing loft, through the hatch in bedroom 3, has uneven, displaced wool around two redundant cold-water tanks (A71–A77); the small loft, reached through a very small hatch, is patchy (A78). There is no roofing felt: the tiles sit on the original timber boarding (A57, A61), so the roof spaces are well ventilated but cold. I saw no water staining on the rafters, and the timbers looked in reasonable condition where visible, though they are worth a roofer checking further; there are old bees' nests. In the rear-wing loft daylight shows where a length of timber is missing (A75).

The assessment enters 100 mm in both lofts, for which the method's assumed U-value is 0.4 W/m²K; at 270 mm it assumes 0.16 W/m²K (RdSAP 10 defaults (BRE, 2026, Table 16) [3]). Where boards crush the insulation the real figure is worse than the default.

What I recommend (Stage 2). Once the roofer has checked the timbers, refixed the tiles and confirmed there are no leaks into the lofts: lift the chipboard, lay mineral wool between the joists and a second layer across them to a total of 270 mm, and re-lay storage boards on loft legs tall enough to clear the new insulation so nothing presses on it; keep a clear air path at the eaves so the roof boarding stays ventilated; drain and remove the redundant tanks or lag them if they are still connected; insulate the three hatches with 100 mm of rigid board and fit compression seals; seal the gaps round the ceiling roses and pipes. Mineral wool: £450–£850 [12]; raised boarding: £500–£1,500 [12].



Main loft: chipboard storage floor over the joists, loose wool at the edges.
A52 · 12:20



Main loft, at the board edge: tape in old mineral wool, about 100 mm and compressed.
A67 · 12:22



Rear-wing loft: displaced wool round a redundant cold-water tank.
A71 · 12:25



Rear-wing loft: uneven wool against the tape; the depth varies across the joists.
A77 · 12:25



Rear-wing loft, roof slope: the gap where a timber appears to be missing; to be checked by a roofer before insulating.
A75 · 12:25



Small loft: the very small hatch to the third roof space.
A78 · 12:24

The ground floor

The main building's floor is suspended timber: pine boards on joists over a void ventilated by air bricks (A4, A38). Under the board lifted in the dining room the void, the joists and lagged pipes are visible, with no insulation (A213, A214); the joists I could see were dry and sound, consistent with the 1987 treatment. It is a small space, and from the one lifted board I could not tell whether it is deep enough to insulate from below; an insulation contractor should be able to confirm that. The method's assumed

U-value for this floor is 0.83 W/m²K; Approved Document L asks 0.25 W/m²K of a renovated floor [2]. The kitchen floor in the rear wing is solid, entered as uninsulated with an assumed 1.23 W/m²K; it is small and would only be insulated in a kitchen refit.

What I recommend. Keep every air brick clear. Filling the gaps between the boards with rubber strips or filler helps for a while, but the fillers tend to work loose and fall out as the boards move, so treat it as a temporary measure. Carpet with a thermal underlay would suit the front living room and stops the draught through the boards for good. If an insulation contractor confirms the void is deep enough, or the boards are ever lifted, insulate between the joists: mineral wool or wood-fibre batts of at least 100 mm on netting, with an airtight membrane over the joists taped to the walls and a vapour-open layer beneath, the pipes lagged and kept on the warm side, and a check for rot as the work goes (Stage 3). £1,200–£2,300 from above [15]. The boards in the living room and hall are a finished floor, so lifting them is the main cost and disruption; if the void is deep enough to work in, insulating from below costs less.



Dining room, lifted board: suspended floor with pipework below.
A213 · 12:30



Dining room, floor void: joist, valves and lagged pipes; no insulation.
A214 · 12:30

Air brick, side wall: ventilates the void; never block it.
A38 · 12:44

The walls

Main building: filled cavity. About 300 mm overall (A179), filled with mineral fibre by CavityTech in October 2005 under BBA certificate 89/2294, with a CIGA 25-year guarantee to August 2030 (A243–A245). I drilled a mortar joint in two elevations and looked in with the borescope: the first (B1–B20) showed fibre against both leaves with some bare brick faces and mortar snots on the wall ties, and small voids (B6); the second (B21–B31) showed fibre filling the cavity, with small air pockets (B28). No wetness or bridging debris was seen. The method assumes 0.7 W/m²K for a filled cavity of this age [3], which suits what I saw; a patchy fill is a little worse than that. No further work is needed now; register your guarantee with CIGA directly, as the installer is believed dissolved.

The downstairs closet (the former tank cupboard). The closet has a small glazed panel with loose insulation fibre behind it (A210); coats there occasionally smell musty and a leather jacket grew white mould (your brief), the same cold-and-still pattern as the wardrobes. Keep it less full and leave the door ajar on dry days.

Rear wing: solid brick. About 260 mm overall, one brick plus plaster, uninsulated, brick below and render above. The method assumes 1.7 W/m²K (RdSAP 10 defaults (BRE, 2026, Table 13), solid brick 200–280 mm) [3]; on the day the walls were dry (meter 152–162) but the paint failure in bedroom 3 and the bathroom shows how cold they run. Approved Document L asks 0.30 W/m²K of an existing wall that is insulated internally or externally [2]: 1.7 to 0.30 is an 82% cut in heat loss through the wing's walls ((1.7 − 0.30) ÷ 1.7). The assessment recommends solid wall insulation. It is worth considering, but on a solid wall it has to be done carefully, and the risks below decide which way it should go.



Bathroom window reveal, rear wing: the tape across the opening gives about 260 mm – a solid wall.
A158 · 11:58



Side door reveal, main building: about 300 mm – the cavity wall.
A179 · 12:10

PHOTO WITHHELD FOR PRIVACY

Cavity fill invoice, 2005: mineral fibre, BBA 89/2294, with the CIGA guarantee.
A245 · 12:57

PHOTO WITHHELD FOR PRIVACY

Borescope point, side wall: probe in a drilled mortar joint, made good afterwards.
A48 · 12:50



Cavity, first elevation: brick face and mortar snot beside patchy fibre; a small void.
B6 · 12:45



Cavity, second elevation: mineral fibre filling the cavity with small air pockets; dry.
B28 · 12:52

Insulating the rear wing: inside or outside?

	INTERNAL WALL INSULATION (ROOM BY ROOM)	EXTERNAL WALL INSULATION (THE WHOLE WING AT ONCE)
What it involves	Insulated plasterboard (about 50 mm PIR or wood fibre) with a vapour control layer, fixed to the outside walls room by room; sockets, radiators, sills and, in wet rooms, sanitaryware and tiling moved and refitted.	Insulation boards with a render finish fixed to the outside of the wing on both floors; sills, downpipes and the soil pipe extended, the eaves detail checked; scaffold.

Effect on the rooms	Loses about 60–70 mm on each outside wall; noticeable in a 4.67 m² bedroom and a small bathroom.	None inside; the rooms can stay in use.
Moisture risk	Higher. The brick becomes colder and dries more slowly; a gap in the vapour control layer lets room air condense behind the board; highest in a bathroom or kitchen.	Lower, if the details are right. The brick stays warm; the risks are at the eaves, sills, reveals and ground level, and from a render that is not breathable.
Thermal bridging	Junctions with the floors, ceilings, window reveals and the main building stay cold unless the insulation is returned.	Continuous over the floor edges; the junction with the main building and the window reveals need detailing.
Planning and consents	Building Regulations notification (a renovated thermal element); planning unlikely to be needed.	Building Regulations notification. Planning may be needed because it changes the wing's appearance; I cannot confirm this, so check with the local council planning team or the Planning Portal [28].
Cost	£2,000–£3,400 upstairs; £1,800–£3,100 for the kitchen [14]	£6,000–£9,500 [14]
Suitability here	**Possible, but difficult in the bathroom and kitchen and in a small bedroom; something to think about with your builder.**	**Worth considering for the solid-wall wing, subject to planning and to a builder's view of the details.**

The risks to weigh

Internal insulation, especially in wet rooms. Insulating a solid wall on the inside changes how it handles moisture: the brick, now on the cold side, gets colder and wetter and dries mainly to the outside, and moist room air that gets behind the insulation through gaps, sockets or fixings can condense on the cold brick, unseen [27]. Bathrooms and kitchens make the most vapour, so the risk is greatest there, and their walls also carry tiles, basins, cupboards and pipes whose fixings puncture the vapour control layer. The cold junctions at floors, ceilings and window reveals stay as mould spots unless the insulation is returned. Floor joists built into the solid wall can be left colder and damper at their ends. These risks can be managed, with a continuous, taped vapour control layer, a condensation-risk check of the build-up to BS 5250, moisture-tolerant boards and good extract ventilation, but they make internal insulation in wet rooms a careful, specified job rather than a quick one.

External insulation. Outside, the brick is kept warm, which is lower-risk for moisture, but the details are where it goes wrong: the eaves and verges may need extending so

rain clears the new face; sills, reveals and pipes must be extended and sealed; the system must stop above ground level and clear of the air bricks; the junction with the cavity-walled main building and the boundary needs careful finishing; and the render must be breathable and kept in repair, because a cracked render lets water in behind the boards [27]. It also changes the look of the house, which may need planning permission.

Which approach suits this house, and in which rooms, can only really be settled with a builder who has looked at the walls, ideally together with the bathroom refurbishment. Either way the repairs and the ventilation come first.

U-values: now, the Regulations, and after the plan

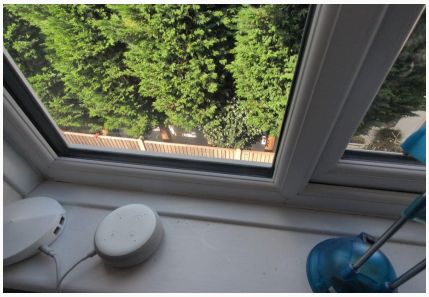
ELEMENT	NOW (ASSUMED BY THE METHOD)	WHAT I FOUND	BUILDING REGULATIONS (RENOVATED ELEMENT)	AFTER THE PLAN (INDICATIVE)
Walls, main (filled cavity)	0.7 W/m²K (RdSAP 10 defaults (BRE, 2026, Table 6)) [3]	Fill present and dry, patchy in places (B6, B28)	0.55 for cavity fill [2]	Unchanged
Walls, rear wing (solid brick)	1.7 W/m²K (RdSAP 10 defaults (BRE, 2026, Table 13)) [3]	260 mm, dry, cold-surface paint failure	0.30 [2]	0.3 W/m²K [2]
Roofs (both lofts)	0.4 W/m²K at 100 mm (RdSAP 10 defaults (BRE, 2026, Table 16)) [3]	About 100 mm, compressed under boards	0.16 [2]	0.16 W/m²K at 270 mm [3]
Floor, main (suspended timber)	0.83 W/m²K (assumed in the assessment)	Uninsulated, gaps between boards	0.25 [2]	0.25 W/m²K [2]
Floor, rear wing (solid)	1.23 W/m²K (assumed in the assessment)	Not seen under the laminate	0.25 [2]	Unchanged (only at a kitchen refit)
Windows	2.7 W/m²K for eight units of unknown date; 2 for bedroom 3 (assessment window table)	Pre-2001 units; one 2016 low-E unit	1.4 for a replacement window (Table 4.2) [2]	Unchanged in this plan

All the "now" U-values are assumed: the assessment uses the method's default U-values for the construction and age ("U-value known: No"), not measured values. I have not changed or recalculated them, and the table comments on them from what I found. On these figures the plan cuts the fabric heat loss shown in chart C5 from about 261 W/K to about 155 W/K, about 40% ((261 − 155) ÷ 261), if

the rear wing and floor are insulated, before any saving from draught-proofing, which the chart does not show.

08 Windows and doors

All nine windows are white uPVC double glazing. Eight predate your purchase in 2001; without a date stamp or FENSA record they are entered as of unknown date, for which the assessment uses 2.7 W/m²K. Bedroom 3's window was replaced in 2016 by Meadvale Windows with 28 mm sealed units and soft-coat low-E glass, and a trickle vent (A247); the assessment enters it as 2002–2021 at 2 W/m²K. One kitchen pane and one French-door pane were reglazed in 2024. The bedroom 2 window collects condensation on its lower panes in winter (your brief) and the bathroom glass was wet at midday (A156) – both signs of humid air rather than failed units. No FENSA certificate was found for the address in the pre-visit search; the 2016 window may have been certified through another scheme. The front door (2021) and the French doors are the draught paths you describe: adjust the keeps, replace worn seals and fit a brush strip to the French-door threshold (Stage 1, within £120–£260).



Bedroom 3 window, 2016: low-E units with a trickle vent – keep it open.
A136 · 11:56

Dining-room French doors: the draught you feel in winter; no trickle vent.
A202 · 12:07

Front door, 2021: the paper test showed it closing loosely on its seal.
A176 · 12:03

- **When the older windows reach the end of their life** (misting units or failing hinges), replace them with units of 1.4 W/m²K or better (Approved Document L Table 4.2) [2], warm-edge spacers, and trickle vents of at least the existing size and no less than 8,000 mm² in a habitable room [1]; fitted by a FENSA or CERTASS installer who self-certifies the work, with the frame sealed to the wall on the inside and the cavity closed at the reveals.
- Until then, the trickle vents in section 06 and the door seals are the only window work in this plan.

09 Roof and walls: keeping the fabric dry

These are the repairs to deal with first. Only when the house is weathertight, with the roof sound, no leaks into the lofts and the outside walls and timber protected, is it worth spending on the improvements in the rest of this report: insulation laid under a leak, or fixed to a wall that is letting in water, traps the water instead of keeping the heat in. Items are tagged ACT NOW (repair before any insulation goes in), PLAN (at the redecoration) or MONITOR (check each winter; no work now). The drone was flown to see the roofs and chimney, which cannot be seen from the ground; its frames are used here where they settle a finding, and the rest are held on file.

ELEMENT	WHAT I SAW	ACTION AND WHO	PRIORITY
Roof tiles	Slipped and missing tiles at the rear eaves (A28); heavy moss on the north-facing hips of the rear-wing roof (D17–D21)	Roofer to replace the missing tiles, refix the slipped ones and remove the moss by hand (no pressure washing)	ACT NOW
Leaks into the lofts	No water staining seen on the rafters I could reach, but the main loft is largely boarded and the lofts have no felt under the tiles	Roofer to check the lofts for leaks while the tiles are repaired; repair any before the lofts are insulated	ACT NOW
Roof timbers	In reasonable condition where seen; daylight where a length of timber appears to be missing in the rear-wing loft (A75; you have seen it)	Roofer to check the timbers further and repair the rear-wing gap before the loft is insulated	ACT NOW
Gutters and downpipes	Gutters not cleared (your brief); moss washing off the rear roof (D17–D21)	Clear the gutters and downpipes £150–£230 [24]	ACT NOW
Render and wall paint	Hairline cracks in the roughcast on the side elevation (A9); weathered paint	Decorator to fill the cracks and repaint with a breathable masonry paint, in the spring redecoration	ACT NOW
Fascias, soffits and bargeboards	Weathered paint on the bargeboards of the front gable (A10) and the fascias and soffits; some timber looks soft	Decorator to probe, replace soft lengths and repaint, in the spring redecoration	ACT NOW

Chimney stack, pots and flues	Four pots with terminals on the stack; lead flashing renewed in 2015/2018; lichen on the stack top (D15). The living-room flue appears open; which pots are capped, and whether any flue has been swept, is not known	A HETAS or NACS sweep to sweep and check the flues in use before the next fire, and confirm which pots are capped	ACT NOW
Pointing	Missing mortar on the side wall, some of it patched	Repoint in lime mortar where joints are open, at the redecoration	PLAN
Air bricks	Clear at the front and sides (A4, A38); one on the neighbour's side looks like a drain grille	Keep clear; never cover with planters or paving	MONITOR
Gully at the rear	Downpipe discharging into an open gully by the wall (A34)	Keep the grid clear of leaves	MONITOR

PHOTO WITHHELD FOR PRIVACY

Front elevation from the drone: roughcast first floor over brick; no open cracks at this level.
D3 · 12:34



Chimney stack on the ridge, from above: four pots with terminals, and lead flashing on the slope; which flues are capped cannot be confirmed from the air.
D15 · 12:35



Main roof, south-east slope: plain clay tiles with lichen; the slope that would take solar panels.
D13 · 12:35

PHOTO WITHHELD FOR PRIVACY

Rear-wing hipped roof: heavy moss on the hips and the north-facing slope, which holds water against the tiles.
D20 · 12:36

Rear eaves: slipped and missing tiles above the gutter line.
A28 · 12:39

Side elevation render: hairline cracking in the roughcast.
A9 · 12:32

Front gable: weathered bargeboards and timber detailing.
A10 · 12:32

The flat roof seen beside the house in the drone frames is on a neighbouring building; it is noted because it adjoins the side passage, and ownership and responsibility for any repair should be confirmed before any work is arranged. Any other defect on the outside of the house is outside the scope of this report; a builder or roofer can advise at the redecoration (section 16).

10 Heating, hot water and low-carbon options

The system you have

A Worcester Greenstar 32CDi Compact ErP condensing combi boiler, GC 47-406-79 (A239), installed in 2021 in the kitchen wall cupboard (A236), serviced every year and guaranteed to 2031; the assessment uses its product-database efficiency of 89.2%. It heats nine radiators and the hot water on demand. The controls are a programmer, a room thermostat and TRVs: Hive provides the programmer and thermostat from a receiver in the kitchen (A231), and TRVs are on five of the nine radiators – bedrooms 1–3, the landing and the hall – with none in the living room, dining room or kitchen (your radiator list).



Boiler data label: Greenstar 32CDi Compact ErP, GC 47-406-79.
A239 · 12:09



Hive receiver, kitchen: heating and hot-water channels.
A231 · 12:06



Bedroom 1 radiator: one of the five with a TRV.
A107 · 12:02

What to do now

- **Fit TRVs to the living-room, dining-room and kitchen radiators (Stage 1).** £300–£500 [21], then balance the system. Leave one radiator without a TRV only where the thermostat sits.

- **Ask about the flow temperature at the next service (no cost).** A condensing boiler is most efficient when the water returning to it is cool, so the engineer can set the radiator flow temperature to the lowest setting that still heats every room on a cold day.
- **Keep the annual service**, which keeps the guarantee to 2031.

Chimneys and flues

The assessment counts four open chimneys and enters the secondary heating as an open fire in a grate, mainly because of the living-room fire, which you use (A191). The dining room also has an open fireplace (A205), used rarely, and the bedroom 1 and 2 fireplaces are closed off at the throat with bagged material (A103, A117).

My advice. I would not seal the chimneys permanently. Open flues lose heat, but they also ventilate the house, which matters here while the humidity is above the healthy range. You already use chimney balloons, and that is the right approach: use them in the flues that are not in use during the winter, and take them out for the rest of the year so the chimneys can ventilate the rooms. The living-room flue appears open, but I could not confirm which pots are capped or whether any of the flues has been swept, so have a HETAS or NACS sweep sweep and check the flues you use before the next fire (section 09), and keep a carbon monoxide alarm in any room with a working fireplace. If you would like to keep fires, a closed stove in the living room loses far less heat up the flue than an open grate; that is a choice for later and outside the costed plan.



Living-room fire: an open grate, in use.
A191 · 12:04



Dining-room fireplace: a cast-iron open fireplace, used rarely.
A205 · 12:08



Bedroom 2 fireplace: the throat closed off with bagged material.
A117 · 11:59

A heat pump, and what must come first

A heat pump is a sensible replacement when the boiler reaches the end of its life, not before: the boiler is five years old and guaranteed to 2031. Before then the house needs the repairs and the draught, loft, rear-wing and floor measures, so that the heat pump is sized on the house's reduced heat loss, the radiators can run cooler, and the running cost stays below the gas cost. The heat pump would also need a hot-water cylinder, for which the installer will need to find a cupboard; an outdoor unit could sit in the garden on the rear wing's south-west side, away from the neighbours' windows.

An installer must carry out a room-by-room heat-loss calculation; a heat pump costs £8,000–£11,000 before the £7,500 Boiler Upgrade Scheme grant [22][8] (section 15).

Solar PV and a battery

No panels are fitted (drone). The main roof's south-east slope (D13) is unshaded at the ridge and could take a system of about 3 kW; the rear-wing hips are small and mossy. The assessment recommends PV. It suits you now that the hob and oven are electric. A 3 kW system costs about £5,500–£7,000 [23]; I place it in Stage 4, after the roof repairs, because the fabric measures do more for your gas bill.

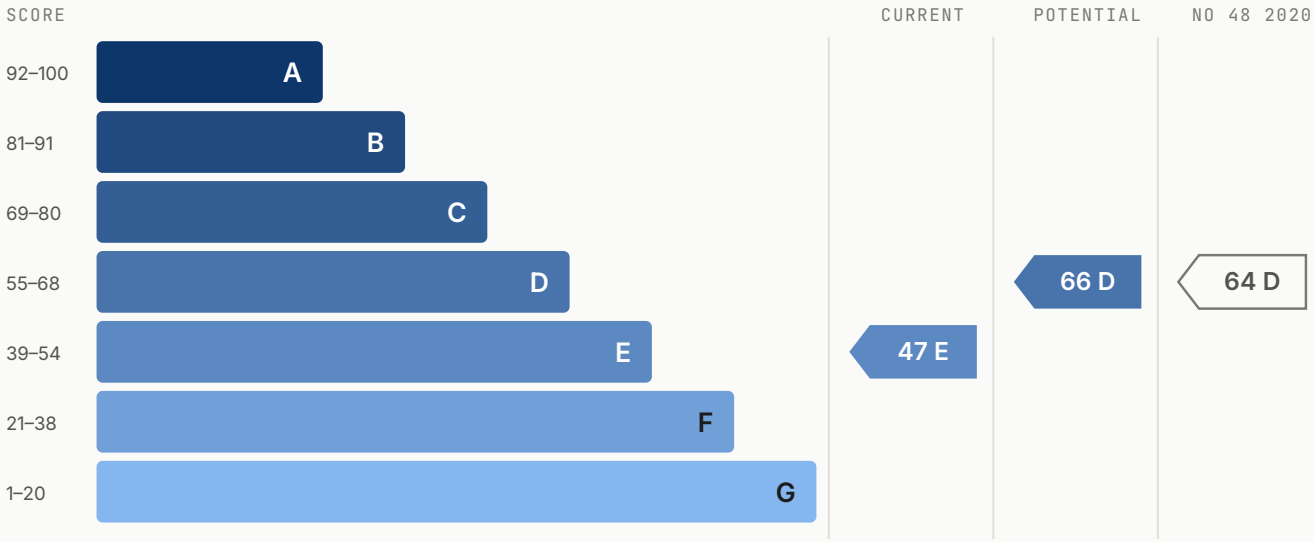
11 Energy performance: the baseline

The figures below come from the RdSAP energy assessment I made on the day of the visit, using the government's RdSAP method. It has not been lodged as an Energy Performance Certificate, but it can be if you wish. The assessment rates the house at **47 (E)** now, with a potential of 66 (D) if its recommended measures are carried out.

What the assessment measures, and what it does not. The method models the house's energy use for a standard household, heating pattern and fuel price, from the dimensions, construction, heating and lighting entered. It uses default U-values for walls, roofs and floors whose performance is not documented, it counts open chimneys and fans rather than measuring air leakage, and it does not see moisture, mould, ventilation rates or how you live. Its running costs and savings are worked out at fixed fuel prices set more than a year ago, so take the £ figures with a pinch of salt: they show roughly what to expect and how measures compare, not what you will pay. It does not predict your bills, which is why section 02 explains your actual gas use separately; your energy supplier's records are the best guide to your actual use and cost.

ENERGY EFFICIENCY RATING	ENVIRONMENTAL (CO_2) RATING	ESTIMATED RUNNING COST	CO_2 EMISSIONS
47 E	**51 E**	**£1,788/yr**	**4.4 t/yr**
Potential 66 D	Potential 62 D	standard occupancy	floor area 85.6 m²

Where the house sits on the rating scale

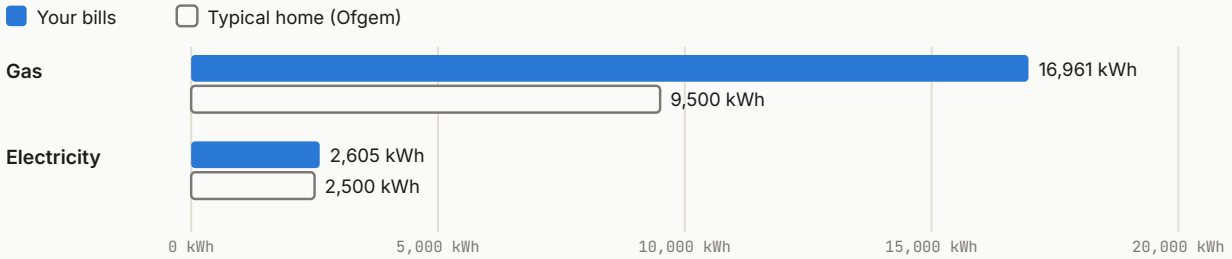


The energy efficiency rating now and its potential. Current 47 (band E) and potential 66 (band D), from the assessment's summary page; the outline pointer is No a neighbouring house, rated 64 (band D) on the EPC register, registered 8 June 2020 – a detached house of the same age with a 250 mm loft.

Estimated running costs, and your actual bills

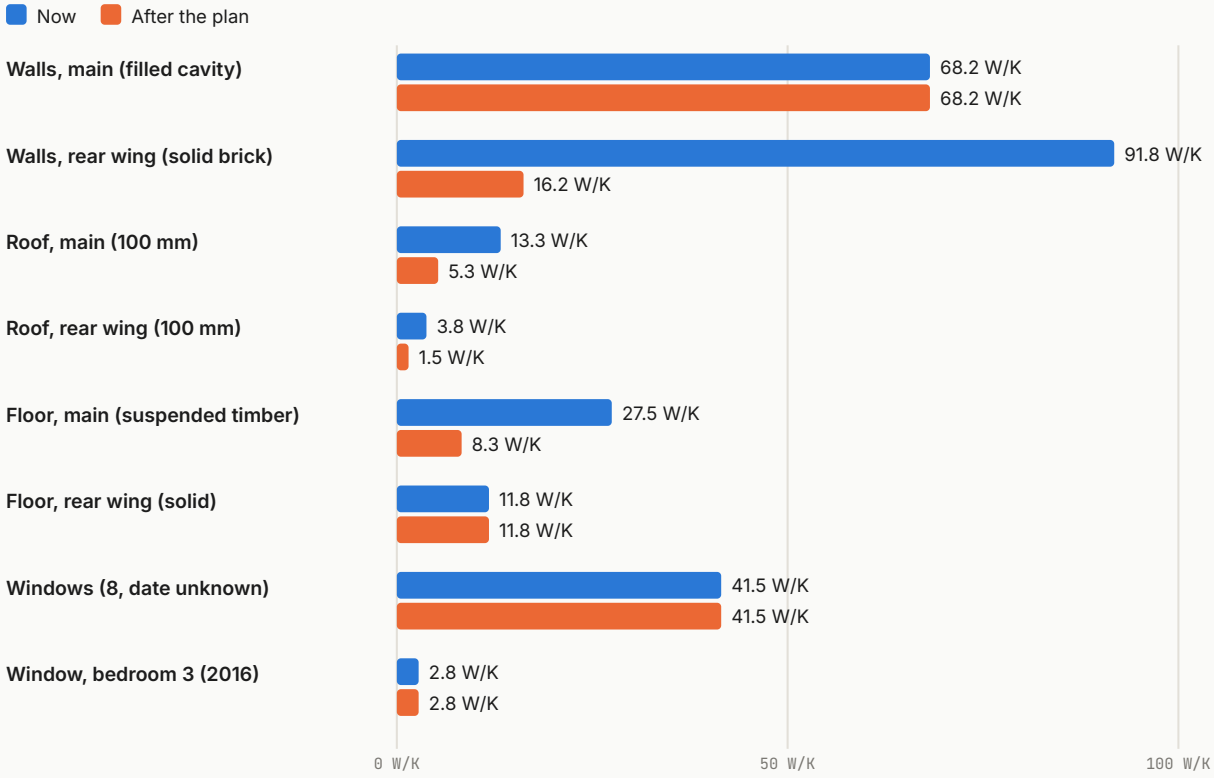
The assessment's total fuel bill is **£1,788 a year** for standard occupancy and its fixed prices; treat it as an indication only. Your own bills come to about £2,495 a year (£1,405 gas and £1,090 electricity, your brief). Actual bills include standing charges and every other use of electricity, and reflect your heating hours rather than the method's standard pattern, so the two figures are not expected to match.

Your energy use against a typical home



Your annual gas and electricity use against Ofgem's typical medium values. Your use from your written brief (gas 1 July 2025 – 20 June 2026, electricity 1 October 2025 – 30 September 2026); typical values from 1 July 2026 [7]. Gas is about 79% above typical ((16,961 − 9,500) ÷ 9,500); electricity about 4% above.

Where the heat goes



Indicative fabric heat loss by element, now and after the plan. Areas from the dimensions page and the assessment's window table; U-values are the method's assumed defaults (RdSAP 10 defaults (BRE, 2026, Tables 6, 13 and 16)) [3] or the values assumed in the assessment, not measured values; fabric elements only – ventilation, chimneys and draughts are not shown, and the one external door is left out because its area was not entered. The "after" values are the Approved Document L improved values [2] and the method's 270 mm loft value, cited in section 07.

What the terms mean

Energy efficiency rating	A score from 1 to 100+ based on the modelled running cost per square metre; bands A (92+) to G (1–20). Higher is better.
Environmental (CO_2) rating	The same scale for the modelled carbon emissions.
Potential rating	The rating after the measures the method recommends for this house (here mainly solid-wall and floor insulation and solar PV), applied in its fixed order.
Estimated costs	The method's running costs for standard occupancy and fixed prices set more than a year ago; useful for comparing, not a forecast of your bills.

How this compares with the register

No certificate exists for a neighbouring house on the EPC register (searched 23 October 2026), so this assessment is the first for the house. The nearest comparable house is No 48, rated **64 (D)**, registered 8 June 2020: also detached and of the same age with a filled cavity, with a 250 mm loft and only a small flat-roofed extension. Its

better rating probably owes much to the deeper loft insulation and to having no solid-brick wing.

The assessment's recommendations, and how this plan treats them

RECOMMENDED BY THE ASSESSMENT	MY VIEW FOR THIS HOUSE
Solid wall insulation (recommended)	Worth considering for the rear wing, inside or outside, after the repairs and with your builder's advice on the risks (section 07, Stage 3).
Floor insulation, suspended floor (recommended)	Yes, if an insulation contractor confirms the void allows it; carpet with thermal underlay in the living room meanwhile (Stage 3).
Solar PV (recommended)	Possible later, after the roof repairs and the fabric measures (Stage 4).
Loft insulation ("increase too small" to be recommended)	Yes, regardless. The method assumes a full 100 mm; the insulation is compressed under boards, and the musty loft and cold bedroom ceilings argue for it, once the roof is repaired (Stage 2).
Draught-proofing ("already installed")	Partly. The windows have seals, but the doors, floors and open flues still leak: door seals in Stage 1, your chimney balloons in winter, and the floor as above.
Heating controls ("already installed")	Partly. Three radiators lack TRVs; Stage 1.
Insulated doors ("increase too small")	Not now; the front door is new.

12 Improvement options

The options are grouped by type, in the order they should be tackled: the repairs, then ventilation and moisture, then draughts, insulation, heating and controls, and finally renewables. The stage shows where each falls in the plan in section 13.

About the costs. Every cost here is indicative: a published range at the date of this report, from the numbered sources in section 17 (national unless marked), scaled to this house for budgeting. Actual prices will vary with the contractor, access, the specification chosen, what is found once work opens up, and changes in prices over time. They are not quotes; obtain three written quotes on the same specification before committing. Savings are published typical figures for standard occupancy, not scaled to your use – with gas use about 1.8 times typical, your savings from the fabric measures are likely to be higher. Comfort and energy benefit: ●●● high · ●●○ medium · ●○○ low · ○○○ none.

A. Repairs to do first

MEASURE	STAGE	INDICATIVE COST	DISRUPTION	WHY
Roofer: replace missing and refix slipped tiles, remove moss, check the lofts for leaks and check the timbers (including the missing rear-wing length)	0	Obtain a quote	Low	The roof must be sound before the lofts are insulated.
Clear the gutters and downpipes	0	£150–£230 [24]	Low	Keeps the walls dry.
Repair and repaint the render, fascias, soffits and bargeboards (breathable masonry paint; soft timber replaced)	0	Obtain a quote	Medium (scaffold)	Protects the walls and timber before any insulation.
Sweep and check the flues in use	0	Obtain a quote	Low	Safety; confirms which pots are capped.

B. Ventilation and moisture

MEASURE	SUITABLE? (STAGE)	COMFORT / ENERGY	INDICATIVE COST	SAVING / YR	DISRUPTION	NOTES AND RISKS
Ducted cooker hood (30 l/s to outside)	Yes (stage 1)	●●○ comfort ○○○ energy	£200–£400 + hood [19]	–	Low	Removes cooking moisture; core-drill through the solid wall.
Trickle vents (every window without one: living room, bedrooms 1 and 2, kitchen, bathroom)	Yes (stage 1)	●●○ comfort ○○○ energy	£200–£400 [20]	–	Low	Fresh-air path for the extract; mark the equivalent area.
Door undercuts (bedrooms 1 and 2, bathroom)	Yes (stage 1)	●○○ comfort ○○○ energy	£60–£120 (my estimate, £20–£40 a door)	–	Low	10 mm above the floor finish.
Bathroom dMEV fan	With the bathroom refit (stage 3)	●●○ comfort ○○○ energy	£250–£450 [18]	–	In the refit	8 l/s continuous, 15 l/s boost, near the shower.

PIV unit	No	–	Not costed	–	–	Pushes moist air into the cold wing and the unfelted roof.

C. Draughts

MEASURE	SUITABLE? (STAGE)	COMFORT / ENERGY	INDICATIVE COST	SAVING / YR	DISRUPTION	NOTES AND RISKS
Door seals and keeps (front, back and French doors), **and filling the floorboard gaps**	Yes (stage 1)	●●○ comfort ●●○ energy	£120–£260 [16]	about £70–£90 [17]	Low	Floor fillers tend to fall out: a temporary measure. Keep the air bricks open.
Carpet with thermal underlay (front living room)	Yes (stage 1)	●●○ comfort ●○○ energy	Not costed (depends on the carpet)	no reliable figure	Low	Stops the draught through the boards for good.
Chimney balloons in the unused flues, winter only	Yes – you already use them	●●○ comfort ●○○ energy	None (already owned)	–	None	Take them out outside winter so the chimneys ventilate the house; do not seal the chimneys permanently.

D. Insulation

MEASURE	SUITABLE? (STAGE)	COMFORT / ENERGY	INDICATIVE COST	SAVING / YR	DISRUPTION	NOTES AND RISKS
Loft top-up to 270 mm (both lofts, hatches)	Yes, after the roof repairs (stage 2)	●●○ comfort ●○○ energy	£450–£850 [12]	about £20–£30 [13]	Low	Roofer to check the timbers and leaks first.
Raised storage boarding (main loft)	Yes (stage 2)	○○○ comfort ●○○ energy	£500–£1,500 [12]	within the loft top-up	Low	Without it the new insulation is crushed.

Suspended floor insulation (main ground floor)	If the void allows (stage 3)	●●○ comfort ●●○ energy	£1,200–£2,300 from above [15]	about £60–£120 [15]	High	A small void; an insulation contractor to confirm it can be done. Air bricks kept open.
Internal wall insulation, rear wing upstairs (bedroom 3, bathroom)	Consider, with your builder (stage 3)	●●● comfort ●●○ energy	£2,000–£3,400 [14]	no reliable figure	Medium	Moisture risk, highest in the bathroom; continuous vapour control layer; loses floor area (section 07).
Internal wall insulation, kitchen	Consider, at a kitchen refit (stage 4)	●●○ comfort ●●○ energy	£1,800–£3,100 [14]	no reliable figure	In the refit	Wet-room risks as above.
External wall insulation, rear wing	Consider, with your builder (stage 3)	●●● comfort ●●○ energy	£6,000–£9,500 [14]	no reliable figure	Medium	Alternative to both internal options; details at eaves and sills; planning to be checked (section 07).
Cavity wall insulation	Already done	–	–	–	–	2005 fill, present and dry; register the guarantee with CIGA.
Extracting the cavity fill	No	–	Not costed	–	–	The fill is dry and doing its job.

E. Heating, controls and windows

MEASURE	SUITABLE? (STAGE)	COMFORT / ENERGY	INDICATIVE COST	SAVING / YR	DISRUPTION	NOTES AND RISKS

TRVs (living room, dining room, kitchen)	Yes (stage 1)	●○○ comfort ●○○ energy	£300–£500 [21]	no reliable figure	Low	Then balance the radiators.
Air source heat pump	Later, if at all (stage 4)	●○○ comfort ●●● energy	£8,000–£11,000 before grant [22] [8]	no reliable figure	High	At boiler end of life, after the fabric measures.
Replacement windows	Later, if at all (stage 4)	●○○ comfort ●○○ energy	Not costed	–	Medium	At the end of the units' life; 1.4 W/m²K with vents.

F. Renewables

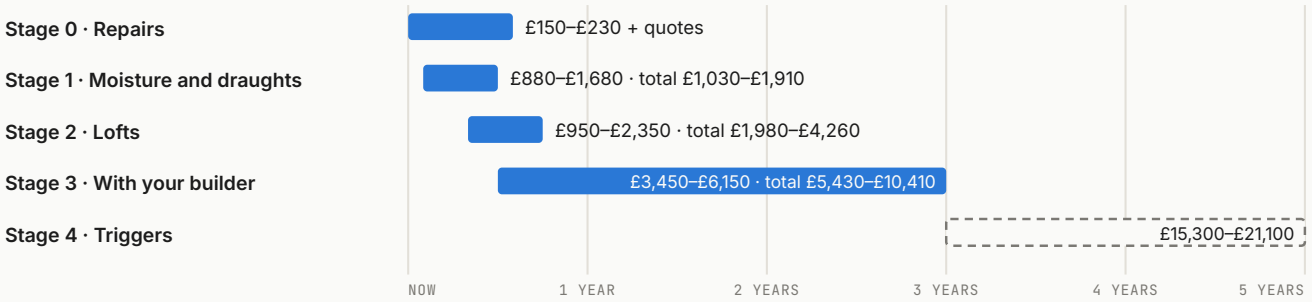
MEASURE	SUITABLE? (STAGE)	COMFORT / ENERGY	INDICATIVE COST	SAVING / YR	DISRUPTION	NOTES AND RISKS
Solar PV, about 3 kW	Later, if at all (stage 4)	○○○ comfort ●●○ energy	£5,500–£7,000 [23]	no reliable figure	Low	South-east slope; after the roof repairs.

13 Your plan: short term and medium term

The plan follows one principle: repairs first, so the house is weathertight; then keep it dry and ventilated; then stop the draughts and insulate; and only then change how it is heated. Each stage makes the next one safe: insulation laid under a leaking roof or against a wet wall traps the water, insulation without ventilation moves condensation onto the next coldest surface, and a heat pump bought before the insulation is oversized for the house it will heat.

Your plans and the triggers they set

YOUR PLAN (AS YOU DESCRIBED IT)	WHEN	WHAT TO FOLD IN	SECTION AND STAGE
External redecoration	Spring 2027	Render repairs and breathable paint; fascias, soffits and bargeboards; if a scaffold goes up, the tiles, gutters and moss	09, Stage 0
Bathroom refurbishment, with the fan moved over the shower	Early 2027	dMEV fan; trickle vent; door undercut; look behind the tiles for mould; decide with the builder whether and how to insulate the outside walls	06 and 07, Stage 3
Staying in the house for the long term	–	The ten-year plan, with any floor insulation in a summer when the rooms can be cleared	07, Stage 3
A kitchen refit (not planned yet)	When it comes	Consider insulating the kitchen walls (with the wet-room risks in mind); a ducted hood on the new layout	07, Stage 4
The boiler reaching the end of its life	After the 2031 guarantee	A heat pump sized on the improved house	10, Stage 4



The plan over five years. Stages 0–3 on a month axis with each stage's indicative cost and the running total; Stage 4 starts at its triggers and runs beyond the axis. The repairs in Stage 0 need quotes, so only the gutter clearance is included in the totals. Cost ranges from costs.json and the sources in section 17.

Short term: the next six months

Stage 0 NOW TO SPRING 2027 · REPAIRS FIRST

MEASURE	WHO	WHEN	INDICATIVE COST	EFFECT
Replace missing and refix slipped tiles, remove the moss, check the lofts for leaks, check the roof timbers and repair the missing rear-wing length	Roofer	November	Obtain a quote	Roof sound to insulate under
Clear the gutters and downpipes	Gutter cleaner	November	£150–£230	Walls kept dry
Sweep and check the flues in use; confirm which pots are capped; CO alarm in rooms with a working fireplace	HETAS / NACS sweep	Before the next fire	Obtain a quote	Safety
Repair the render and repaint with breathable masonry paint; repair and paint the fascias, soffits and bargeboards; repoint open joints in lime	Decorator; builder	Spring 2027	Obtain a quote	Walls and timber protected
Stage subtotal			**£150–£230 plus the quotes**	

Stage 1 THIS WINTER · MOISTURE AND DRAUGHTS

These are small, low-risk jobs that deal with the mould, so they can go ahead while the repairs are being arranged.

MEASURE	WHO	WHEN	INDICATIVE COST	SAVING / YR	EFFECT
Ducted cooker hood (30 l/s or more)	Electrician	Within 3 months	£200–£400 + hood	–	Kitchen moisture removed
Trickle vents to the five windows without them	Window installer	Within 3 months	£200–£400	–	Fresh-air path in every room
Trim the bedroom 1, bedroom 2 and bathroom doors to a 10 mm gap	Carpenter	Within 3 months	£60–£120	–	Air reaches the fans
Door seals and keeps (front, back, French doors); fill the floorboard gaps as a temporary measure; consider carpet with thermal underlay in the living room	You or a handyman	Now	£120–£260	about £70–£90	Warmer hall and living room
TRVs on the living-room, dining-room and kitchen radiators; balance the system	Gas Safe engineer	At the service	£300–£500	–	Room-by-room control
Chimney balloons in the unused flues for the winter (out again in spring); keep the bedroom 3 vent open; a hygrometer on the landing	You	Now	No cost	–	Fewer draughts, less damp
Stage subtotal			**£880–£1,680**	**up to about £70–£90 a year**	**running total £1,030–£1,910 plus the Stage 0 quotes**

Stage 2 ONCE THE ROOF IS REPAIRED · LOFTS

MEASURE	WHO	WHEN	INDICATIVE COST	SAVING / YR	EFFECT
Loft top-up to 270 mm in all three lofts, hatches insulated and sealed	Insulation installer	After the roofer	£450–£850	about £20–£30	Warmer bedroom ceilings
Raised storage boarding in the main loft	Same installer	With the top-up	£500–£1,500	–	Insulation not crushed
Stage subtotal			**£950–£2,350**	**up to about £20–£30 a year**	**running total £1,980–£4,260 plus the Stage 0 quotes**

Medium term: six months to three years

Stage 3 2027–2029 · WITH YOUR BUILDER, TIED TO THE BATHROOM REFIT

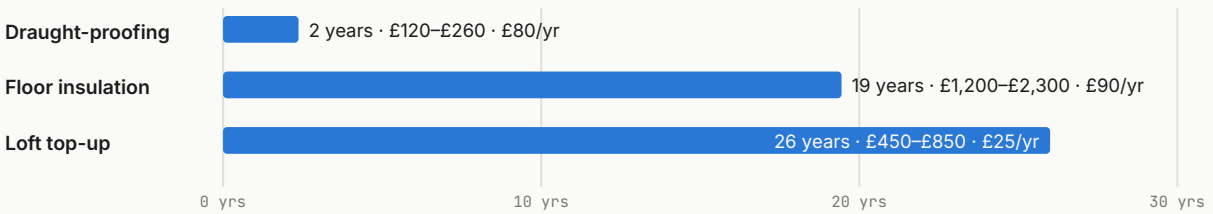
MEASURE	WHO	WHEN	INDICATIVE COST	SAVING / YR	EFFECT
Bathroom dMEV fan, trickle vent and undercut	Bathroom builder; electrician	Early 2027	£250–£450	–	Bathroom mould stops
Decide with your builder whether to insulate the rear wing, and how: internal insulation of bedroom 3 and the bathroom (cost shown), or external insulation of the whole wing (£6,000–£9,500, planning to be checked) – section 07	Builder; insulation installer	Early 2027	£2,000–£3,400	no reliable figure	Bedroom 3 warmer and drier
Insulate the suspended floor between the joists, if an insulation contractor confirms the void allows it	Insulation installer; carpenter	Summer 2027 or 2028	£1,200–£2,300	about £60–£120	Warm floors, no draughts
Stage subtotal (with internal wall insulation)			**£3,450–£6,150**	**up to about £60–£120 a year**	**running total £5,430–£10,410 plus the Stage 0 quotes**

Longer term: three to ten years, at trigger points

Stage 4 AT TRIGGER POINTS · HEATING AND THE BIGGER MEASURES

MEASURE	TRIGGER	INDICATIVE COST
Internal insulation of the kitchen walls, if you choose it (with the wet-room risks in section 07 in mind)	A kitchen refit	£1,800–£3,100
Air source heat pump sized on the improved house, with a hot-water cylinder	The boiler's end of life, after 2031	£8,000–£11,000 before the £7,500 grant
Solar PV, about 3 kW, on the south-east slope	After the roof repairs, when budget allows	£5,500–£7,000
Stage subtotal		**£15,300–£21,100**

All costs in this plan are indicative and will vary: they are published ranges scaled to this house, not quotes, and the final price depends on the contractor, the specification and what is found once work starts. The repairs in Stage 0 cannot be priced without a roofer and decorator seeing the work. Obtain three written quotes for each measure.



Simple payback: cost midpoint ÷ typical saving. Only measures with both a sourced cost and a sourced saving: draught-proofing (saving £70–£90, detached house [17]), floor insulation (£60–£120 [15]) and the loft top-up from 100 to 270 mm (£20–£30 [13]). Typical savings for standard use; yours are likely to be higher. The ventilation measures have no energy payback – they are there to stop the mould.

Quick wins and the end state

Quick wins this winter: the door seals (£120–£260 with the floor gaps), your chimney balloons in the unused flues, keeping the bedroom 3 vent open and asking about the boiler flow temperature at the next service – for a saving of about £70–£90 a year in a typical detached house [17] and a less draughty hall and living room.

The end state: after Stages 0–3 the roof and walls are sound, the bathroom and kitchen clear their moisture, the lofts are insulated, the draughts are reduced, and – if your builder confirms the approach – the rear wing and the main floor are insulated. The fabric heat loss shown in chart C5 then falls by about 40%. The total indicative spend for Stages 0–3 is £5,430–£10,410, plus the roof and redecoration quotes. The method's potential rating is 66 (D), which assumes its own recommended measures including solar PV.

> **Sequencing matters.** Carrying out measures in the wrong order is a common cause of retrofit problems. The order above avoids four of them: insulating over a defect (a loft insulated under a leak or a missing timber, or a wall insulated while its render lets in water); a measure that does not work in isolation (sealing up the house without the hood, fan and trickle vents in place raises the humidity); a measure that would have to be undone (insulating the bathroom walls after it has been tiled); and a measure that is the wrong solution where another would do better (a heat pump sized on today's house, or a PIV unit where the wet rooms need extract). Stage savings are shown "up to about £x a year before interaction between measures": savings from separate measures overlap and do not simply add up.

How you will know it is working

- Use the meter readings on the day as your baseline: gas 45009 cubic feet (A217) and electricity 42523 kWh (A218), both on Thursday 22 October 2026.
- Read the gas meter every Monday morning through the winter and compare the weekly use with the same weeks last year on your Octopus account; your supplier's records are the best measure of what the work saves.
- Keep a hygrometer on the landing and in bedroom 3: below 60% RH most of the time, and no condensation on the bedroom 3 walls by spring.
- Ask the installer to re-test the new hood and bathroom fan on completion: 30 l/s and 15 l/s (or 8 l/s continuous) as a minimum.

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14 What to avoid

- **Do not insulate before the repairs.** Insulation laid under a leaking roof or fixed to a wall that lets in water traps the water and hides the damage.
- **Do not block the air bricks** or fill the void under the floor; the joists stay dry only because the void is ventilated.
- **Do not seal up the house before you ventilate.** Draught-proofing and insulation without the hood, fan and trickle vents would push the humidity higher.

- **Do not seal the chimneys permanently.** They ventilate the house; use your chimney balloons in winter only and take them out for the rest of the year.
- **Do not use a cement render or plastic masonry paint** on the rear wing's solid brick or the roughcast: they trap moisture in the wall. Use a breathable paint.
- **Please avoid spray foam** on the underside of the roof boarding: it hides the timber, traps moisture and can make the house hard to sell or mortgage.
- **Do not fit internal wall insulation without a specification**, above all in the bathroom or kitchen: it needs a continuous vapour control layer, taped at the joints and the edges, and a condensation-risk check of the build-up to BS 5250, the British Standard for managing moisture in buildings, or the condensation simply moves behind the board (section 07).
- **Do not buy a heat pump sized on today's house**; it would be oversized and less efficient once the insulation is in.

15 Funding

- **Boiler Upgrade Scheme.** £7,500 towards an air source heat pump in England and Wales, for an owner replacing fossil-fuel heating, checked on gov.uk on 23 October 2026 [8] [9]; the page sets no EPC condition and gives no end date. The scheme may change before your boiler is due for replacement.
- **VAT.** Installing insulation, draught stripping, heating controls, heat pumps, solar panels and batteries is zero-rated until 31 March 2027, and 5% from 1 April 2027 (VAT Notice 708/6) [10]. The loft, floor and wall work in Stages 2–3 is worth timing before April 2027 where you can; the rate may change.
- **Local and national grants.** The Great British Insulation Scheme ended on 31 March 2026 [11]. I found no current grant for which a two-adult owner-occupied household is clearly eligible. Local schemes open and close through the year, so check the the local council and the county council websites before you commission the larger works.
- **Smart meter and tariff.** You have a smart electricity meter but a mechanical gas meter; ask Octopus to fit a smart gas meter so that weekly use is recorded automatically (section 13). With an electric hob and oven and, later, solar PV or a heat pump, a time-of-use tariff may suit you.

16 Finding the right trades

WORK	REGISTER TO USE	SEARCH

Insulation (loft, floor, walls) and draught-proofing	TrustMark – https://www.trustmark.org.uk/homeowner/find-a-tradesperson	"loft insulation", "internal wall insulation", "underfloor insulation" near [postcode]
Trickle vents and future windows	FENSA – https://www.fensa.org.uk/find-installers; CERTASS – https://certass.co.uk/	"window installer" near [postcode]
Boiler, TRVs and flow temperature	Gas Safe – https://www.gassaferegister.co.uk/find-an-engineer/	"heating engineer" near [postcode]
Cooker hood and bathroom fan wiring	NICEIC – https://www.niceic.com/find-a-contractor; NAPIT – https://search.napit.org.uk/	"electrician" near [postcode]
Heat pump and solar PV	MCS – https://mcscertified.com/find-an-installer/	"air source heat pump", "solar PV" near [postcode]

The MCS register showed one certified installer within about 25 miles among the entries it displayed on 23 October 2026; the other registers cannot be searched by a script, so the table gives the searches to run.

BUSINESS	TOWN	TRADE	REGISTER AND NUMBER	CHECKED ON
Your Energy Your Way CIC	Egham (TW20 8RX)	MCS-certified installer; the technologies were not shown	MCS NAP-12624 – https://mcscertified.com/find-an-installer/	23 October 2026

Other professionals. For the chimney sweep, search HETAS or NACS for your postcode; for a roofer, TrustMark; for any crack that a builder thinks structural, a chartered structural engineer through the Institution of Structural Engineers' directory. I have not verified any firm on those registers.

I have no connection with any business listed. A firm shown under MCS, FENSA, Gas Safe or NICEIC was verified on that register only; whether it is also TrustMark-registered is not something I have checked unless its TrustMark licence number is given, so check the TrustMark registration yourself at https://www.trustmark.org.uk/homeowner/find-a-tradesperson before instructing anyone (funded work must use a TrustMark-registered installer working to PAS 2030). Obtain three written quotes on the same specification; send the quotes to me and I will review them against this report.

17 Next steps, assessor, sources and limitations

Next steps

1. **Read the report and call me with any questions.** A follow-up conversation is included in the fee.
2. **Book the repairs now (Stage 0).** A roofer for the missing and slipped tiles, any leaks into the lofts and the roof timbers; a gutter clearance; a sweep for the flues you use; and the decorator's quote for the walls, fascias, soffits and bargeboards in the spring.
3. **This winter, the moisture and draught measures (Stage 1).** The cooker hood, trickle vents, door trimming, door seals and TRVs; your chimney balloons in the unused flues for the winter only; keep the bedroom 3 vent open.
4. **Once the roofer has confirmed the roof is sound, the lofts (Stage 2).**
5. **Before the bathroom is priced, talk to your builder (Stage 3).** Share sections 06 and 07 with them, and decide together whether and how to insulate the rear wing; check the planning position for external insulation with the local council or the Planning Portal; ask an insulation contractor whether the floor void can be insulated.
6. **Get three written quotes for each measure** on the specification in this report, and send them to me if you would like them reviewed against it.
7. **If you would like the energy assessment lodged** as an Energy Performance Certificate, tell me and I will lodge it.

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Sources

Cost figures are national unless marked: the TRV figure is from local listings, the gutter-cleaning figure is the London and South East rate, and the door-trimming figure is my own estimate; every other range is a national figure scaled to this house. All costs are indicative and may vary. Each source was read on the date shown.

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Limitations

This report is advisory. It is not a design, a specification, a structural assessment, a building survey or a condition report, and it is not an Energy Performance Certificate; the energy assessment has not been lodged, though it can be on request. The inspection was visual and non-invasive except for the borescope holes drilled in mortar joints in two elevations (made good) and one floorboard lifted in the dining room; the floor void beyond that board, the wall behind the bathroom tiles, the rear-wing floor and the inside of the flues were not seen, and the lofts were viewed from the boarded and accessible areas only. Readings reflect the conditions on the day; moisture readings are comparative; U-values are the method's assumed defaults, not measurements; the energy assessment's costs use fixed fuel prices and are indicative only; costs are indicative ranges from the sources listed, national unless marked, and are not quotes. Prepared for the client under the Terms and Conditions of Engagement sent on 8 October 2026 and accepted on 15 October 2026; no third party may rely on it.